Graph Theory

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Chapter Summary

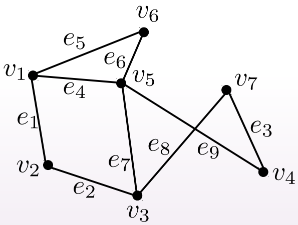
1. Graph introduction
2. Graph terminology and special types of graphs
3. Graph representations
4. Graph isomorphism
5. Connectivity
6. Euler and Hamiltonian graphs
7. Planar graphs

Graph Theory: Introduction

Graph: Consists of a set (non-empty) of vertices/nodes V and a set of edges E. such that each edge is associated with one or two vertices.

Denoted by $G(V, E)$

The vertices associated with an edge are called **end points.**



A graph G

Vertex-set $V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$

Edge-set $E = \{e_1 = \{v_1, v_2\}, e_2 = \{v_2, v_3\}, e_3, e_4, e_5, e_6, e_7, e_8, e_9\}$

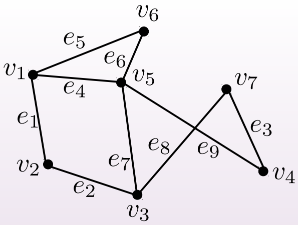
Graph Theory

V : nodes/vertices/points E : edges/arcs/links

Edges is a relation defined on V

Graph size:

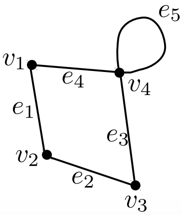
number of nodes $n = |V|$ number of edges $m = |E|$



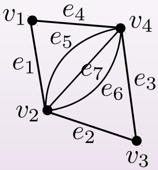
$n = 7$, $m = 9$

Types of graphs

Loop/self-loop: Edges whose both end points are the same.



Multiedges: there are more than one edge between two end points.



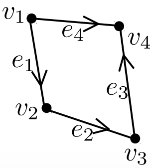
Types of graphs

Simple graph: A graph is simple if every edge is associated/connected with two different vertices. In other words, the graph contains no self-loop and multiedges.

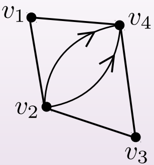
Multigraph: Simple graph + multiple edges (multiedges)

Pseudograph: Simple graph + multiple edges (multiedges) + self-loops

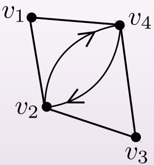
Directed graph: A graph $G(V, E)$ with a set of vertex V and a set of edges E, where the edges are an ordered pair of elements of V (directed edges)



Multiedges:



Multiedges



Not multiedges

Types of graphs

Directed multigraph: Simple directed graph + directed multiedges

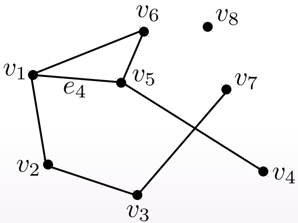
Let $e = (u, v)$ be a directed edge.

u : initial vertex

v : end/terminal vertex

Degree: Undirected graphs

Degree: degree of a vertex is the number of edges associated/incident with that vertex.

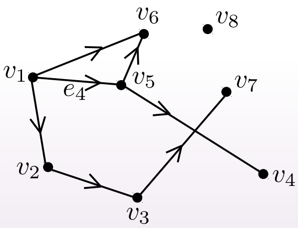


$deg(v_5) = 4$	
$deg(v_4) = 1$	pendant vertex
$deg(v_8) = 0$	isolated vertex vertex

Degree: directed graphs

Two types of degree: In-degree ($deg^{-}(v)$) and out-degree ($deg^{+}(v)$)

- In-degree ($deg^{-}(v)$): number of edges directed towards a vertex
- Out-degree ($deg^{+}(v)$) number of edges directed outwards from a vertex



$deg^{-}(v_5) = 1$
$deg^{+}(v_5) = 2$

The Handshaking Theorem

The Handshaking Theorem: Let $G(V, E)$ be an undirected graph then, $\sum_{v \in V} deg(v) = 2|E|$

Proof: Each edge contributes 2 in the degree count one for each end point.

Another Theorem

Theorem: An undirected graph has an even number of odd degree vertices.

Proof:

- We have $\sum_{v \in V} deg(v) = 2|E|$
- V_{odd} : set of odd-degree vertices
- V_{even} : set of even-degree vertices
- We have $\sum_{v \in V_{even}} deg(v) + \sum_{v \in V_{odd}} deg(v) = 2|E|$
- $\sum_{v \in V_{even}} deg(v)$ and $2|E|$ are even then $\sum_{v \in V_{odd}} deg(v)$ must be even.
- Since each vertex in V_{odd} is odd-degree and their degree sum is even, implies that there are even number of vertices.

Theorem on directed graph

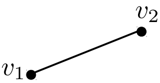
Theorem: In a directed graph $G(V, E)$,
$\sum_{v \in V} deg^{-}(v) + \sum_{v \in V} deg^{+}(v) = |E|$

Proof: Each edge contributes 1 in $deg^{-}(v)$ and 1 in $deg^{+}(v)$

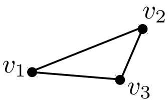
Types of graphs: Regular graph

Regular graph: A simple graph is called regular if every vertex is of the same degree.

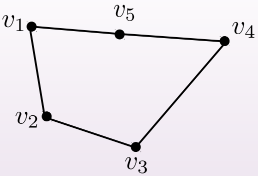
n-regular graph: degree of each vertex is n



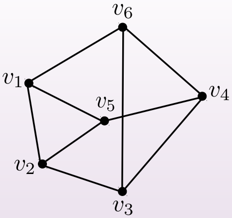
1-regular



2-regular



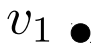
2-regular



3-regular

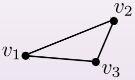
Complete graph: A simple graph is called complete if every pair of vertices is connected by an edge.

K_n: complete graph of size n, i.e., n vertices.

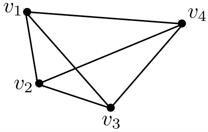


K_1

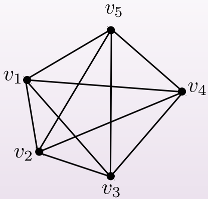
K_2



K_3



K_4



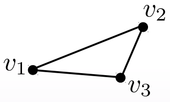
K_5

Types of graphs: Cycle graph

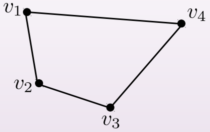
Cycle graph: A cycle consisting of $n \geq 3$ vertices $v_1, v_2, \ldots, v_n$ and edges $\{v_1, v_2\}$, $\{v_2, v_3\}$, ..., $\{v_{n-1}, v_n\}$, $\{v_n, v_1\}$.

C_n: cycle graph of size n, that is, n vertices.

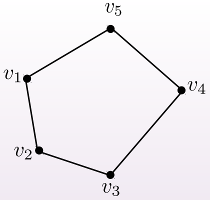
It is a 2-regular graph.



C_3



C_4

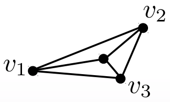


C_5

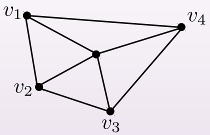
Types of graphs: Wheel graph

Wheel graph: A wheel graph consisting of $n \geq 3$ vertices $v_1, v_2, \ldots, v_n$ is a cycle graph C_n plus one vertex v and each vertex in C_n is connected by an edge to v.

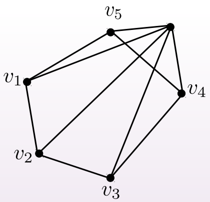
W_n: wheel graph of size $n+1$, that is, $n+1$ vertices.



W_3



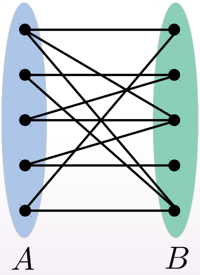
W_4



W_5

Bipartite graphs

Bipartite graph: Simple graph $G(V, E)$ whose vertex-set V is partitioned into two sets A and B ($A \cap B = \emptyset$) and the edges connect vertices of A with the vertices of B.



Relation with cycle graph:

C_5 is bipartite? No

C_6 is bipartite? Yes

C_n is bipartite for $n \geq 4$ is even

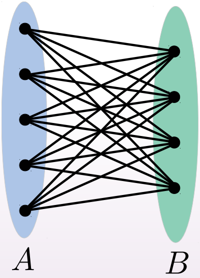
Bipartite graph

Result: A simple graph G is bipartite if and only if it is two-colorable.
Proof:

Complete bipartite graph

Complete bipartite graph: A graph is bipartite plus complete.

$K_{m,n}$: Complete graph with $|A| = m$ and $|B| = n$.

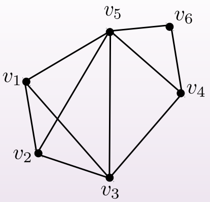


$K_{5,4}$

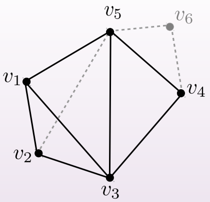
Subgraphs

Subgraph: A subgraph of a graph $G(V, E)$ is a graph $H(V', E')$ where $V' \subseteq V$ and $E' \subseteq E$.

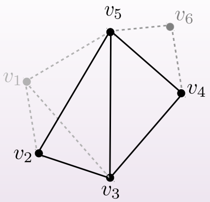
Induced subgraph: A subgraph of a graph $G(V, E)$ is an induced subgraph $H(V', E')$ where the edge-set E' contains an edge in E if and only if both endpoints of this edge are in E'.



A graph



A subgraph



An induced subgraph

Complementary graph

Complementary graph: The complementary graph $\overline{G}$ of a simple graph G has the same vertices as G.

Two vertices are adjacent in G if and only if they are not adjacent in $\overline{G}$.

Example:

Degree seq, Havel-Hakimi theorem

Degree sequence: The degree sequence of a simple graph G is the sequence of the degrees of the vertices in G non-decreasing order.

$(d_1, d_2, \ldots, d_n)$ where $d_1 \geq d_2 \geq \ldots \geq d_n$.

Example:

Graphic: A sequence $(d_1, d_2, \ldots, d_n)$ is called **graphic** if it is the degree sequence of a simple graph.

Havel-Hakimi Theorem: Consider the following two sequences and assume sequence (1) is in non-increasing order.

1. $s, t_1, t_2, \ldots, t_s, d_1, d_2, \ldots, d_n$
2. $t_1 - 1, t_2 - 1, \ldots, t_s - 1, d_1, d_2, \ldots, d_n$.

The sequence (1) is graphic if and only if sequence (2) is graphic

Example: $(5, 3, 3, 3, 3, 2, 2, 2, 1, 1, 1)$ is graphic if and only if $(2, 2, 2, 2, 1, 2, 2, 1, 1, 1)$ is graphic.

Graph representations

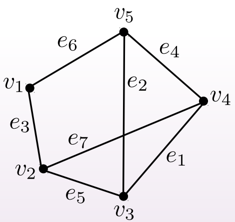
- Adjacency matrix representation
- Adjacency list representation
- Incidence matrix representation

Adjacency matrix representation

It is a two-dimensional $(0,1)$-matrix M of $n \times n$ size where n is the number of vertices.

$M = [m_{ij}]_{n\times n}$, where

$$m_{ij} = \begin{cases} 1, & \text{if } v_i \text{ and } v_j \text{ is connected by an edge} \\ 0, & \text{otherwise} \end{cases}$$



	v_1	v_2	v_3	v_4	v_5
v_1	0	1	0	0	1
v_2	1	0	1	1	0
v_3	0	1	0	1	1
v_4	0	1	1	0	1
v_5	1	0	1	1	0

Weighted graph:

$$m_{ij} = \begin{cases} w_{ij}, & \text{if } v_i \text{ and } v_j \text{ is connected by an edge and its weight is } w_{ij} \\ 0, & \text{otherwise} \end{cases}$$

Adjacency matrix representation

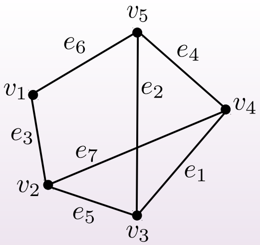
Properties of Adjacency Matrix Representation

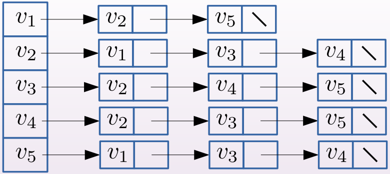
- Memory required
  - $\Theta(n^2)$, independent on the number of edges in G
- Preferred when
  - The graph is dense: m is close to n^2
  - need to quickly determine if there is an edge between two vertices
- Time to list all vertices adjacent to u:
  - $\Theta(n)$
- Time to determine if (u, v) belongs to E:
  - $\Theta(1)$

Adjacency list representation

Adjacency list representation of $G = (V, E)$

- An array of n lists, one for each vertex in V
- Each list $Adj[u]$ contains all the vertices v such that there is an edge between u and v
  - $Adj[u]$ contains the vertices adjacent to u (in arbitrary order)
- Can be used for both directed and undirected graphs





Weighted graph: Extra field in each node represent the weight.

Adjacency list representation

Properties of Adjacency List Representation

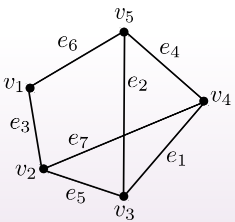
- Memory required
  - $\Theta(m + n)$
- Preferred when
  - the graph is sparse: $m << n^2$
- Disadvantage
  - no quick way to determine whether there is an edge between node u and v
  - Time to determine if (u, v) exists: $O(degree(u))$
- Time to list all vertices adjacent to u:
  - $\Theta(degree(u))$

Incidence matrix representation

It is a two-dimensional $(0,1)$-matrix M of $n \times m$ size where n is the number of vertices.

$M = [m_{ij}]_{n\times m}$, where

$$m_{ij} = \begin{cases} 1, & \text{if } e_j \text{ is incident on } v_i \\ 0, & \text{otherwise} \end{cases}$$

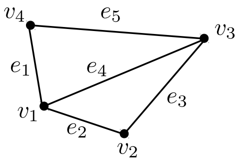


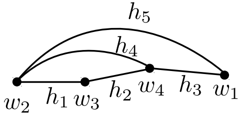
	e_1	e_2	e_3	e_4	e_5	e_6	e_7
v_1	0	0	1	0	0	1	0
v_2	0	0	1	0	1	0	1
v_3	1	1	0	0	1	0	0
v_4	1	0	0	1	0	0	1
v_5	0	1	0	1	0	1	0

Weighted graph:

$$m_{ij} = \begin{cases} w_{ij}, & \text{if } e_j \text{ is incident on } v_i \text{ and weight of } e_j \text{ is } w_{ij} \\ 0, & \text{otherwise} \end{cases}$$

Graph isomorphism





Are thee graphs identical?

$v_1 \equiv w_4$

$v_2 \equiv w_3$

$v_3 \equiv w_2$

$v_4 \equiv w_1$

$\{v_1, v_2\} \equiv \{w_4, w_3\}$

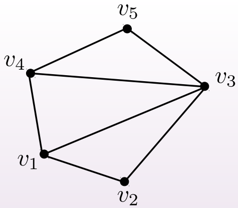
$\{v_2, v_3\} \equiv \{w_3, w_2\}$

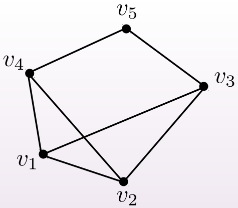
$\vdots$

Graph isomorphism

Graph isomorphism: Let $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$ be two graphs. G_1 and G_2 are isomorphic if there is a function f from G_1 to G_2 that is one to one and onto such that u_1 and w_1 are adjacent in G_1 iff $f(u_1)$ and $f(w_1)$ are adjacent in G_2, for all $u_1, w_1 \in G_1$.

Example:

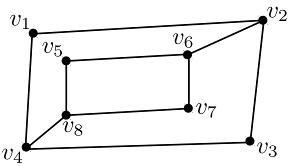


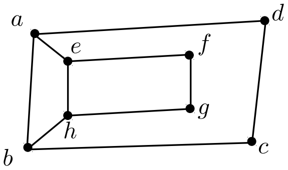


Not isomorphic as degree differs

Graph isomorphism

Example:

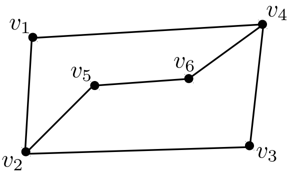


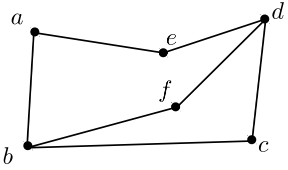


- consider v_1 that is a degree 2 vertex.
- In G_1, v_1 is connected to v_2 and v_4 and both are the degree 3 vertex.
- $f(v_1)$ must be a degree 2 vertex in G_2. It is one of $\{c, d, f, g\}$.
- $f(v_1)$ must connect to the 2 degree 3 vertex.
- a contradiction, because none of m{c,d,f,g} connects the 2 degree 3 vertex.
- So, G_1 and G_2 are not isomorphic.

Graph isomorphism

Example:





- $f(v_1)$ is either c or f
- $f(v_2)$ is either b or d
- $f(v_3)$ is either f or c
- $f(v_4)$ is either d or b
- $f(v_5)$ is either e or a
- $f(v_6)$ is either a or e

They are isomorphic

Walk, path, cycle, circuit

Walk: An alternating sequence of vertices and edges in a graph. $v_1, e_1, v_2, e_2, v_3, e_3, \ldots, v_n$ where end points of e_i are v_i and v_{i-1} for $1 \leq i \leq n-1$

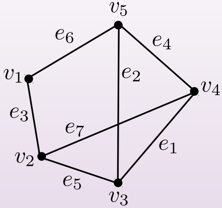
Trail: A walk with no edge is repeated.

Path: A walk with no vertex is repeated.

Closed walk: $v_n = v_1$

Circuit/closed trail: $v_n = v_1$ in a trail

Cycle/closed path: $v_n = v_1$ in a path



Walk: $v_1, e_3, v_2, e_7, v_4, e_4, v_5, e_6, v_1, e_3, v_2, e_5, v_3$

Trail: $v_2, e_7, v_4, e_1, v_3, e_5, v_2, e_3, v_1$

Path: $v_2, e_5, v_3, e_2, v_5, e_4, v_4$

Euler graph

Euler tour: It is a circuit that visits every edge in a graph.

Euler graph: A graph where a Euler tour exists is called a Euler graph.

Hamiltonian cycle: It is a cycle that visits every vertex in a graph.

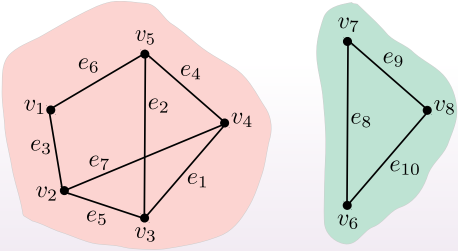
Hamiltonian graph: A graph where a Hamiltonian cycle exists is called a Hamiltonian graph.

Result: A graph G is Eulerian if and only if it is connected and the degree of each vertex is even.

Connected graph

Connected graph: A graph G is connected if there is a path between each pair of vertices in G.

Connected components: Maximal connected subgraph

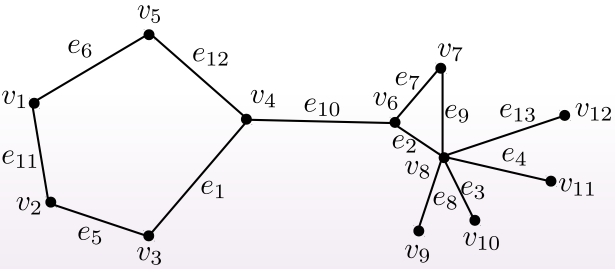


component 1 component 2

Cut vertices and cut edges

Cut vertex: It is a vertex whose removal makes the graph disconnected.

Cut edge: It is an edge whose removal makes the graph disconnected.



Cut vertices: v_4, v_8

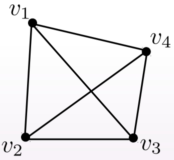
Cut edges: e_{10}, e_8

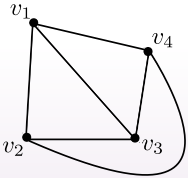
Planar graph

Planar graph: A graph is called planar if it can be drawn in the plane without any edges crossing.

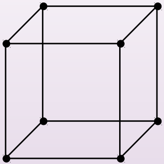
- Crossing of edges is the intersection of the lines or arcs representing them at a point other than their common endpoint.
- Such a drawing is called a planar representation of the graph.

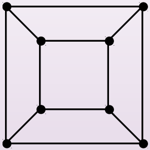
k_4





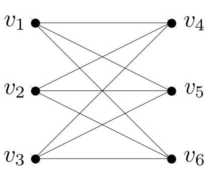
Q_4

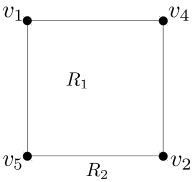


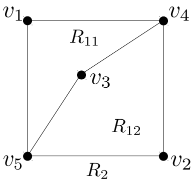


Planar graph

Example: $K_{3,3}$ is not planar.







- v_1 and v_2 must be connected to both v_4 and v_5.
- These four edges form a closed curve that splits the plane into two regions, R_1 and R_2.
- v_3 is in either R_1 or R_2.
- When v_3 is in R_1, the inside of the closed curve, the edges between v_3 and v_4 and between v_3 and v_5 separate R_1 into two subregions, R_{11} and R_{12}.
- There is no way to placev_6 without forcing a crossing.
- A similar argument can be used when v_3 is in R_2 .

Euler's Formula

Theorem: Euler's fromula:: Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of regions in a planar representation of G. Then $v - e + r = 2$.

Corollary: If G is a connected planar simple graph with e edges and v vertices, where $v \geq 3$, then $e \leq 3v - 6$.

Corollary: If G is a connected planar simple graph, then G has a vertex of degree not exceeding five.

Corollary: If a connected planar simple graph has e edges and v vertices with $v \geq 3$ and no circuits of length three, then $e \leq 2v - 4$.

Euler's Formula

Example: Show that K_5 is nonplanar.
Solution:

- K_5 has 5 vertices and 10 edges.
- From Euler's formuls, the inequality $e \leq 3v - 6$ is not satisfied, because $3v - 6 = 9$ and $e = 10$.

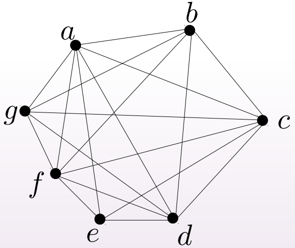
Example: Show that $K_{3,3}$ is nonplanar.
Solution:

- $K_{3,3}$ has no circuits of length three (this is easy to see because it is bipartite),
- $K_{3,3}$ has 6 vertices and 9 edges.
- $2v - 4 = 8$ and $e = 9$.

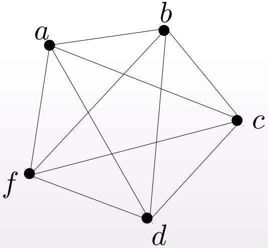
Kuratowski's Theorem

Theorem: A graph is nonplanar if and only if it contains a subgraph homeomorphic to $K_{3,3}$ or K_5.

Example: Show that the following graph is not planar.



The graph G



a subgraph K_5 of G

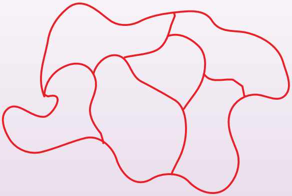
Graph coloring and dual graph

Coloring maps: Color a map such that two regions with a common border are assigned different colors.

Dual graph: Each map can be represented by a graph:

- Each region of the map is represented by a vertex;
- Edges connect two vertices if the regions represented by these vertices have a common border.
- Two regions that touch only at one point are not considered adjacent.

The resulting graph is called the **dual graph** of the map.



Graph coloring and dual graph

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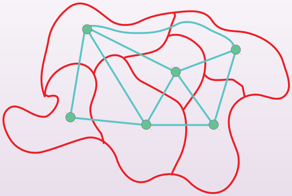

Graph coloring

Coloring: A coloring of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.

Chromatic number: The chromatic number of a graph is the least number of colors needed for a coloring of this graph.
The chromatic number of a graph G is denoted by $\chi(G)$. (Here χ is the Greek letter chi.)

The four color theorem

Theorem: The chromatic number of a planar graph is no greater than four

A few examples

Example: What is the chromatic number of K_n?

Solution:

- Every two vertices of this graph are adjacent.
- No two vertices can be assigned the same color.
- The chromatic number of K_n is n, i.e., $\chi(K_n) = n$.

Example: What is the chromatic number of $K_{m,n}$?

Solution:

- It is a bipartite graph
- The chromatic number of $K_{m,n}$ is 2, i.e., $\chi(K_{m,n}) = 2$.